

J:\noorejara_task6_TENPYTHON DEVELOPER1159.py - C:/Users/Intlj/AppData/Local/Programs/Python/Python311/noorejara_task6_TENPYTHON DEVELOPER1159.py (3.11.5)
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```
target_units = int(input("Enter target units: "))
workers = int(input("Enter number of workers per shift: "))
defect_rate = float(input("Enter defect rate percentage: "))

total_produced = 0
total_defects = 0

print("\nProduction Started...\n")

for shift in range(1, 4):    # 3 shifts
    print(f"Shift {shift}")

    for machine in range(1, 21):    # 20 machines

        produced = workers * 2

        defects = int((produced * defect_rate) / 100)

        good_items = produced - defects

        total_produced += good_items
        total_defects += defects

        print(f"Machine {machine}: Good Items = {good_items}, Defects = {defects}")

    if total_produced >= target_units:
        print("\nTarget Achieved! Production Stopped.")
        break

if total_produced >= target_units:
    break

print("\nProduction Summary")
print("Total Good Items Produced:", total_produced)
print("Total Defective Items:", total_defects)

productivity = total_produced / workers
print("Worker Productivity:", productivity)
```

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```
Python 3.11.5 (tags/v3.11.5:ccc6ba9, Aug 24 2023, 14:38:34) [MSC v.1936 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
- RESTART: C:/Users/Intlj/AppData/Local/Programs/Python/Python311/noorejara_task6_TENPYTHON DEVELOPER1159.py
Enter target units: 200
Enter number of workers per shift: 3
Enter defect rate percentage: 7

Production Started...

Shift 1
Machine 1: Good Items = 6, Defects = 0
Machine 2: Good Items = 6, Defects = 0
Machine 3: Good Items = 6, Defects = 0
Machine 4: Good Items = 6, Defects = 0
Machine 5: Good Items = 6, Defects = 0
Machine 6: Good Items = 6, Defects = 0
Machine 7: Good Items = 6, Defects = 0
Machine 8: Good Items = 6, Defects = 0
Machine 9: Good Items = 6, Defects = 0
Machine 10: Good Items = 6, Defects = 0
Machine 11: Good Items = 6, Defects = 0
Machine 12: Good Items = 6, Defects = 0
Machine 13: Good Items = 6, Defects = 0
Machine 14: Good Items = 6, Defects = 0
Machine 15: Good Items = 6, Defects = 0
Machine 16: Good Items = 6, Defects = 0
```

Machine 16: Good Items = 6, Defects = 0
Machine 17: Good Items = 6, Defects = 0
Machine 18: Good Items = 6, Defects = 0
Machine 19: Good Items = 6, Defects = 0
Machine 20: Good Items = 6, Defects = 0

Shift 2

Machine 1: Good Items = 6, Defects = 0
Machine 2: Good Items = 6, Defects = 0
Machine 3: Good Items = 6, Defects = 0
Machine 4: Good Items = 6, Defects = 0
Machine 5: Good Items = 6, Defects = 0
Machine 6: Good Items = 6, Defects = 0
Machine 7: Good Items = 6, Defects = 0
Machine 8: Good Items = 6, Defects = 0
Machine 9: Good Items = 6, Defects = 0
Machine 10: Good Items = 6, Defects = 0
Machine 11: Good Items = 6, Defects = 0
Machine 12: Good Items = 6, Defects = 0
Machine 13: Good Items = 6, Defects = 0
Machine 14: Good Items = 6, Defects = 0

Target Achieved! Production Stopped.

Production Summary

Total Good Items Produced: 204

Total Defective Items: 0

Worker Productivity: 68.0